21 A spherical water droplet with density 1000 kg m^{-3} and diameter 1.20 μm is suspended in a uniform electric field. The electric field strength is 462 N C^{-1} and is directed downwards. How many excess electrons does it have?

A 1.92×10^{-17} **B** 120 **C** 192 **D** 1.20×10^{11}

Net force = 0
$QE = mg$
$(nq)E = \rho\,(4/3)\,\pi\,r^3\,g$
$n = (\rho\,(4/3)\,\pi\,r^3\,g) / (qE)$
$= (1000\,(4/3)\,\pi\,(0.60\times10^{-6})^3\,g) / (1.6\times10^{-19} \times 462) = 120$

Ans: B